CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

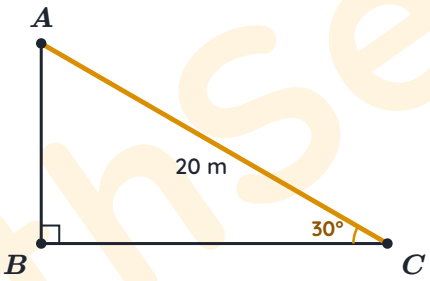
Chapter 9: Some Applications of Trigonometry

Step-by-step solutions to the textbook exercises: 15 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 9.1

Q1 A 20 m rope is stretched tight from the top of a vertical pole to the ground and makes an angle of 30° with the ground. Find the height of the pole.



SOLUTION

Let AB be the pole and AC the rope. In right triangle ABC, $\sin 30^\circ = \dfrac{AB}{AC}$.

$\dfrac{1}{2} = \dfrac{AB}{20}$, so $AB = 10$ m.

Answer: 10 m

Q2 A tree breaks in a storm and its top touches the ground 8 m from the foot of the tree, making an angle of 30° with the ground. Find the height of the tree.

SOLUTION

Let the tree break at a height h, and let the broken part have length l. The standing part, the ground and the broken part form a right triangle with base 8 m.

$\tan 30^\circ = \dfrac{h}{8}$ gives $h = \dfrac{8}{\sqrt{3}}$ m. $\cos 30^\circ = \dfrac{8}{l}$ gives $l = \dfrac{16}{\sqrt{3}}$ m.

Height of the tree $= h + l = \dfrac{24}{\sqrt{3}} = 8\sqrt{3}$ m.

Answer: $8\sqrt{3}$ m (about 13.86 m)

Q3 Two slides are planned in a park: one whose top is 1.5 m high and inclined at 30° to the ground, and a steeper one whose top is 3 m high and inclined at 60°. Find the length of each slide.

SOLUTION

In each case the slide is the hypotenuse and the height is the side opposite the angle, so length $= \dfrac{\text{height}}{\sin\theta}$.

First slide: $\dfrac{1.5}{\sin 30^\circ} = \dfrac{1.5}{1/2} = 3$ m. Second slide: $\dfrac{3}{\sin 60^\circ} = \dfrac{3 \times 2}{\sqrt{3}} = 2\sqrt{3}$ m.

Answer: 3 m and $2\sqrt{3}$ m

Q4 From a spot on level ground 30 m away from the base of a tower, the top of the tower is seen at an elevation of 30°. How tall is the tower?

SOLUTION

$\tan 30^\circ = \dfrac{h}{30}$, so $h = \dfrac{30}{\sqrt{3}} = 10\sqrt{3}$ m.

Answer: $10\sqrt{3}$ m

Q5 A kite flies 60 m above the ground. Its string is tied to a point on the ground and makes an angle of 60° with the ground. Find the length of the string, assuming it is straight.

SOLUTION

$\sin 60^\circ = \dfrac{60}{l}$, so $l = \dfrac{60 \times 2}{\sqrt{3}} = 40\sqrt{3}$ m.

Answer: $40\sqrt{3}$ m

Q6 A boy 1.5 m tall stands some distance from a 30 m tall building. As he walks towards it, the angle of elevation of its top from his eyes increases from 30° to 60°. How far did he walk?

SOLUTION

The top of the building is $30 - 1.5 = 28.5$ m above his eye level.

At 30° his distance from the building is $\dfrac{28.5}{\tan 30^\circ} = 28.5\sqrt{3}$ m. At 60° it is $\dfrac{28.5}{\tan 60^\circ} = \dfrac{28.5}{\sqrt{3}}$ m.

Distance walked $= 28.5\sqrt{3} - \dfrac{28.5}{\sqrt{3}} = \dfrac{28.5 \times 2}{\sqrt{3}} = \dfrac{57}{\sqrt{3}} = 19\sqrt{3}$ m.

Answer: $19\sqrt{3}$ m

Q7 A transmission tower stands on top of a 20 m tall building. From one point on the ground, the bottom of the tower is seen at an elevation of 45° and its top at 60°. Work out the height of the tower.

SOLUTION

Let the point be d m from the building and the tower be h m high.

$\tan 45^\circ = \dfrac{20}{d}$ gives $d = 20$. $\tan 60^\circ = \dfrac{20 + h}{d}$ gives $20 + h = 20\sqrt{3}$.

Answer: $h = 20(\sqrt{3} - 1)$ m (about 14.64 m)

Q8 A 1.6 m tall statue is mounted on a pedestal. An observer on the ground sees the top of the statue at an elevation of 60° and the top of the pedestal at 45°. How high is the pedestal?

SOLUTION

Let the pedestal be h m high and the point be d m away. $\tan 45^\circ = \dfrac{h}{d}$ gives $d = h$.

$\tan 60^\circ = \dfrac{h + 1.6}{h}$ gives $h\sqrt{3} = h + 1.6$, so $h = \dfrac{1.6}{\sqrt{3} - 1} = \dfrac{1.6(\sqrt{3} + 1)}{2} = 0.8(\sqrt{3} + 1)$.

Answer: $0.8(\sqrt{3} + 1)$ **m (about 2.19 m)**

Q9 Standing at the foot of a tower, you see the top of a building at an elevation of 30°. Standing at the foot of the building, you see the top of the tower at 60°. If the tower is 50 m tall, how tall is the building?

SOLUTION

Let the distance between them be d m and the building be h m high.

$\tan 60^\circ = \dfrac{50}{d}$ gives $d = \dfrac{50}{\sqrt{3}}$. $\tan 30^\circ = \dfrac{h}{d}$ gives $h = \dfrac{d}{\sqrt{3}} = \dfrac{50}{3}$.

Answer: $16\dfrac{2}{3}$ **m**

Q10 Two poles of equal height stand on opposite sides of a road 80 m wide. From a point on the road between them, the angles of elevation of their tops are 60° and 30°. Find the height of the poles and the distances of the point from them.

SOLUTION

Let each pole be h m high and the point be x m from the pole seen at 60°, so $(80 - x)$ m from the other.

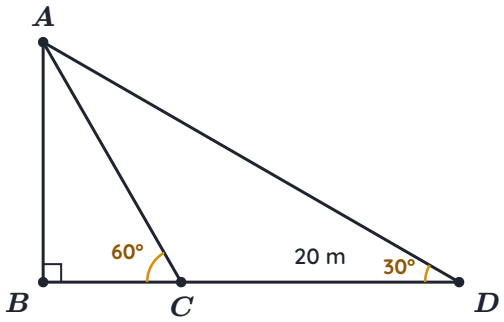
$\tan 60^\circ = \dfrac{h}{x}$ gives $h = \sqrt{3}\,x$. $\tan 30^\circ = \dfrac{h}{80 - x}$ gives $h = \dfrac{80 - x}{\sqrt{3}}$.

So $3x = 80 - x$, $x = 20$ and $h = 20\sqrt{3}$.

Answer: the poles are $20\sqrt{3}$ **m high, and the point is 20 m and 60 m from them**

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Q11 A TV tower stands on one bank of a canal. From the point directly opposite on the other bank the angle of elevation of its top is 60°. From a point 20 m further back on the same line the angle is 30°. Find the height of the tower and the width of the canal.



SOLUTION

Let $AB = h$ be the tower and $BC = x$ the width of the canal, with $CD = 20$ m.

In $\triangle ABC$: $\tan 60^\circ = \dfrac{h}{x}$, so $h = \sqrt{3}\,x$. In $\triangle ABD$: $\tan 30^\circ = \dfrac{h}{x+20}$, so $x + 20 = \sqrt{3}\,h = 3x$.

So $x = 10$ and $h = 10\sqrt{3}$.

Answer: the tower is $10\sqrt{3}$ m high and the canal is 10 m wide

Q12 A building is 7 m tall. From its roof, the top of a cable tower is seen at an elevation of 60° and the base of the tower at a depression of 45°. Find the height of the cable tower.

SOLUTION

Let the horizontal distance be d m. The angle of depression of the foot is 45°, so $\tan 45^\circ = \dfrac{7}{d}$ and $d = 7$.

The part of the tower above the level of the building top is $d\tan 60^\circ = 7\sqrt{3}$ m.

Answer: $7 + 7\sqrt{3} = 7(\sqrt{3}+1)$ m (about 19.12 m)

Q13 From the top of a lighthouse 75 m above sea level, the angles of depression of two ships are 30° and 45°. One ship is exactly behind the other on the same side. Find the distance between the ships.

SOLUTION

The ship seen at 45° is $\dfrac{75}{\tan 45^\circ} = 75$ m from the foot of the lighthouse. The ship seen at 30° is $\dfrac{75}{\tan 30^\circ} = 75\sqrt{3}$ m away.

Answer: $75\sqrt{3} - 75 = 75(\sqrt{3}-1)$ m (about 54.9 m)

Q14 A girl 1.2 m tall sees a balloon moving in a horizontal line at a height of 88.2 m. The angle of elevation of the balloon from her eyes is 60° and, some time later, 30°. Find the distance travelled by the balloon in that time.

SOLUTION

The balloon is $88.2 - 1.2 = 87$ m above her eye level.

Horizontal distance at 60°: $\dfrac{87}{\tan 60^\circ} = \dfrac{87}{\sqrt{3}} = 29\sqrt{3}$ m. At 30°: $\dfrac{87}{\tan 30^\circ} = 87\sqrt{3}$ m.

Answer: $87\sqrt{3} - 29\sqrt{3} = 58\sqrt{3}$ m

Q15 A straight highway leads to the foot of a tower. From the top of the tower a car is seen at an angle of depression of 30°, approaching at a uniform speed. Six seconds later the angle of depression is 60°. How much more time does the car take to reach the foot of the tower?

SOLUTION

Let the tower be h m high. The car is first $\dfrac{h}{\tan 30^\circ} = \sqrt{3}\,h$ away and then $\dfrac{h}{\tan 60^\circ} = \dfrac{h}{\sqrt{3}}$ away.

In 6 seconds it covers $\sqrt{3}\,h - \dfrac{h}{\sqrt{3}} = \dfrac{2h}{\sqrt{3}}$. The remaining distance $\dfrac{h}{\sqrt{3}}$ is half of this, so at the same speed it takes half the time.

Answer: 3 seconds

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